# Deficiency Is a Module

## Channel-Resolved Closure, Vocabulary Granularity, and the Operator Gap on Token Arenas

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## Abstract

$f_H(V) = dim\ V$

*module* D = A/W, and every reader's channel deficit is a Fourier or isotypic coefficient of a single character $\chi_D = \chi_A - \chi_W$; the projector apparatus used to measure them is eliminable. Second, which irreducible types are reachable as deficiencies depends on the granularity of the declaration vocabulary, not on the arena: the same 12-vertex witness realizes a trivial obstruction at five-family resolution and a three-dimensional sign-twisted obstruction at orbit resolution, with opposite channel behaviour. Third, reader saturation is not closure — an exhibited vocabulary has $P_0W = P_0A$ while $W \neq A$ — and the criterion for when saturation does certify closure is a visibility condition that turns out to be Frobenius reciprocity.

Two results concern dynamics. A field on the arena with equivariant local evolution reproduces the module statement exactly: a reader-null sector stays invisible under every admissible linear law, sources the visible channel through the lowest-order equivariant nonlinearity, and is then recoverable from a single visible observation up to sign. Pushing upstream, the arena does not select an evolution operator: equivariance, radius-one locality, self-adjointness and conservation leave a 22-parameter family whose dimension equals the number of edge orbits, and the standard graph Laplacian is the point where all 22 weights are equal. Requiring the transformation price to be generated by an equivariant potential cuts 22 to 7, excludes the Laplacian by a single edge orbit, and leaves the arena connected.

Positioned against the literature, the representation-theoretic content of this paper is recovered rather than new; the direction of arrival, the arena's measured data, and one question — which deficiency types a given vocabulary lattice can reach — are what the work contributes. Full numerical records and non-claims are stated explicitly.

## 1. What Papers I–III Established, and the One Correction

$K = 2$ token arena $A_2$

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$$G = Aut(X) \cong C_2 \times A_4, \quad |G| = 24.$$

Inside the 300-dimensional cochain space, the span of the 3- and 4-cell boundary vectors is the ambient relation module

$$dim\ A = 235, \quad dim\ A^G = 15, \quad m(A) = (15, 4, 14, 4, 24, 36),$$

*f*

> **Correction. (the scope of Papers I–III)** The functional $f_H(V) = \dim V^H$ is the trivial-character coefficient of a complete decomposition, not the whole reading. Every numerical result of Papers I–III is correct as the $\alpha = 1_H$ slice. What was too narrow is the quantifier: an unindexed predicate "this vocabulary closes" omits the reader and the channel. From here every closure claim carries (H, k) for a cyclic reader and (H, α) in general.

That correction is not a repair of the earlier arithmetic. It reorganizes it, and the reorganization has consequences the original framing could not express. Sections 2 through 4 develop them; Sections 6 through 8 follow the same correction into dynamics.

# 2. The Deficiency Module and Its Character

Let $W \subseteq A$ be the span of a declared vocabulary. Papers I–III treated the shortfall as a scalar, $\delta = \dim A - \dim W$. The first result of this paper is that the scalar is a dimension of a structured object and the structure is what answers every reader at once.

## 2.1 Spans are reader-stable, and the channel split is a character

| vocabulary | r | $r_0$ | $r_1$ | $r_0+r_1$ | defect |
|---|---|---|---|---|---|
| disj | 216 | 105 | 111 | 216 | 0 |
| spec4 | 168 | 84 | 84 | 168 | 0 |
| rot4 | 150 | 72 | 78 | 150 | 0 |
| spec3 | 174 | 87 | 87 | 174 | 0 |
| rot3 | 20 | 10 | 10 | 20 | 0 |
| disj+spec4 | 234 | 114 | 120 | 234 | 0 |
| rot3+spec3 | 193 | 97 | 96 | 193 | 0 |
| rot3+spec3+spec4 | 199 | 100 | 99 | 199 | 0 |
| all five | 235 | 115 | 120 | 235 | 0 |

*Table 1. Split defect $r - r_0 - r_1$ across vocabulary unions. Zero in all 31 cases: every span is ρ-invariant, so the channels partition it with no loss. Selected rows shown; the three with $r_1 < r_0$ are the odd-dimensional spans.*

> **Proposition 2.1. (channel ranks from one trace)** For a ρ-invariant span W with $\rho^2 = I$, $r_0(W) - r_1(W) = Tr(\rho \mid W)$, hence $r_0 = (r + Tr)/2$ and $r_1 = (r - Tr)/2$. Computed independently as a trace, this reproduces every one of the 31 measured channel ranks. The explicit projectors are therefore a verification instrument, not the mechanism.

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## 2.2 Deficiency is conserved across channels

$$\delta_0 + \delta_1 = 235 - r = \delta, \quad \delta_0 - \delta_1 = -5 - Tr(\rho \mid W)$$

show that the total shortfall does not change with the channel; the channel decides only where it sits. The vocabulary disj+spec4 is short exactly one dimension in the ambient. That dimension is z-even, so it lies entirely in k = 0, and the k = 1 reader does not see more — the hole is simply not in its channel. "Fails by one dimension" was never channel-free.

> **Theorem 2.2. (the deficiency module)** D = A/W is a G-module with character $\chi_D = \chi_A - \chi_W$. For a cyclic reader H = ⟨g⟩ of order m, the channel deficit is the Fourier coefficient
>
> $$\delta_{H,k}(W) = (1/m)\, \Sigma_j\, \zeta_m^{-jk}\, \chi_D(g^j),$$

## 2.3 Fifty channels, two outcomes

With closure reduced to a character computation, the declaration tree at every cyclic channel follows with no projectors, no per-channel rank and no search. Across all 15 cyclic reader classes and 50 channels, with capacities ranging from 38 to 139:

| quantity | k = 0 | every k ≠ 0 |
|---|---|---|
| closing vocabularies (of 31) | 12 | 15 |
| declaration paths to closure | 82 | 68 |
| branch points | 76 | 66 |
| path lengths | {2:4, 3:18, 4:36, 5:24} | {2:8, 3:12, 4:24, 5:24} |

*Table 2. Channel-conditioned declaration trees. Two outcomes across fifty channels, split on whether the channel is the trivial character. The k = 0 row reproduces the trivial-reader counts of Papers I–III, which is the regression control.*

Only three of the 31 vocabulary subsets change closure status between k = 0 and k = 1, two of them minimal; patching the k = 0 predicate with exactly those two reproduces the k = 1 tree to the last length bucket, and removing them recovers k = 0. The mechanism is therefore an exact predicate-level identity rather than an inference from totals.

# 3. Granularity: the Reachable Deficiency Spectrum

## 3.1 A retracted refutation, and why it belongs in the record

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**Hypothesis 3.1. (equivariant vocabularies)** The character machinery requires $\rho(g)W = W$ for all $g$. The admissible search domain is therefore unions of G-orbits of cells, whose spans are invariant by construction. This is not a technicality: it states what an equivariant vocabulary declaration is.

## 3.2 A $3^-$ obstruction in the same arena

| quantity | value |
|---|---|
| orbits in the union | 33 |
| dim W | 232 (dim A = 235) |
| W is G-invariant | yes, verified |
| dim $W^G$ | 15 = dim $A^G$ |
| dim D | 3 |
| m(D) | $\{3^- : 1\}$ |

*Table 3. An orbit-level vocabulary with full invariant rank and nonzero deficiency. Verified exactly over $\mathbb{Q}$, not modulo a prime.*

| deficiency type | δ at k = 0 | δ at k = 1 | what the invariant reader reports |
|---|---|---|---|
| $D \cong 1^+$ (disj+spec4) | 1 | 0 | sees the hole; k = 1 appears closed |
| $D \cong 3^-$ (33-orbit union) | 0 | 3 | reports closed; k = 1 sees all of it |

*Table 4. Mirror obstructions in one arena, one group, one ambient module, one cell inventory. Only the declaration language differs.*

A second construction, filling every channel but one and preferring orbits that contribute least to the target, isolates $3^-$ with 15 orbits and $m(D) = \{3^- : 2\}$. For the other five irreducibles the same construction closed the target as well, which is a failure of that construction and not a proof of impossibility.

**Definition 3.2. (reachable deficiency spectrum)** For a vocabulary family $\mathcal{V}$ with admissible unions S and spans $W_S \subseteq A$, set $\mathfrak{D}(\mathcal{V}) = \{ [A/W_S] : S \text{ admissible} \}$, with irreducible support $\mathrm{Supp}(\mathcal{V}) = \{ \lambda : \exists S, m_\lambda(A/W_S) > 0 \}$.

Measured on this witness: Supp(five families) = $\{1^+\}$ across all 31 unions, while Supp(90 orbits) $\supseteq \{1^+, 3^-\}$. The arena, the group and the rational irreducibles are unchanged. Only the admissible declaration language changes, so granularity is an algebraic control parameter — refining the vocabulary changes which quotients are reachable.

**Corollary 3.3. (three statements rescoped)** Over the five-family lattice: every nonzero deficiency contains $1^+$; the trivial channel is the hardest, so k = 0 closure implies closure at every $k \neq 0$ (zero counterexamples over 50 channels); and there are exactly two channelwise declaration trees. All three are false over the orbit lattice. They are family-lattice statements, not arena-wide statements.

# 4. Reader Saturation Is Not Closure

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## 4.1 The criterion

> **Theorem 4.1. (vocabulary-relative certification)** Invariant saturation at reader H certifies genuine closure over a vocabulary class $\mathcal{V}$ if and only if dim $\lambda^H > 0$ for every $\lambda \in \mathrm{Supp}(\mathcal{V})$. A reader is not faithful or unfaithful in the abstract; it is certificate-faithful relative to the deficiency types its declaration language can realize.

## 4.2 The diagnostic inversion, which is Frobenius reciprocity

H

| reader | order | obstruction types its invariant channel can see |
|---|---|---|
| trivial | 1 | 6 of 6 |
| an involution | 2 | 6 of 6 |
| order 3 | 3 | 4 of 6 |
| order 4 | 4 | 3–4 of 6 |
| order 6 | 6 | 2 of 6 |
| order 8 | 8 | 2 of 6 |
| order 12 | 12 | 2 of 6 |
| full Aut(X) | 24 | 1 of 6 — only $1^+$ |

*Table 5. Visibility by reader. dim $\lambda^H$ is non-increasing along subgroup containment — zero violations across all 37 containment pairs among the twelve readers of Papers I–III.*

> **Corollary 4.2. (the diagnostic ceiling)** The apparatus of Papers I–III is the invariant channel of the full automorphism reader. It can see exactly one of the six rational obstruction types. It was correct about $1^+$ because $1^+$ is the only type it could ever have found. The most symmetric reader is the blindest.

## 4.3 One asymmetry, two faces

# 5. Position Relative to Established Mathematics

The programme's standard is collision: a framework that produces only its own objects cannot be checked. Accordingly the machinery of Sections 2–4 was audited against the literature, and the result should be stated plainly rather than softened.

| result in this paper | established name | class |
|---|---|---|
| $\chi_D = \chi_A - \chi_W$ | additivity of characters on exact sequences | known |
| $r = r_0 + r_1$, zero split defect | Maschke / complete reducibility | known |
| $r_0 = (r + \mathrm{Tr}\,\rho)/2$ | eigenspace dimensions of an involution | known |
| $\delta_{H,k}$ as a Fourier coefficient | character multiplicity formula | known |
| $\delta_{H,\alpha} = \langle \mathrm{Res}\,\chi_D, \chi_\alpha \rangle$ | orthogonality relations | known |

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| result in this paper | established name | class |
|---|---|---|
| $W^H = A^H \iff D^H = 0$ | exactness of $(-)^H$ in characteristic 0 | known |
| dim $\lambda^H$ non-increasing in H | $\lambda^H \subseteq \lambda^K$ for $K \leq H$ | known |
| the diagnostic inversion (§4.2) | Frobenius reciprocity | known |
| closure factorizes per channel | central idempotents; Wedderburn | known |
| min generators ≥ max$\lambda$ $\lceil m_\lambda s_\lambda / \dim \lambda \rceil$ | minimal generation of a semisimple module | known |
| $2_\pm$ with $\langle\chi,\chi\rangle = 2$ | Schur index; rational vs absolute irreducibility | known |
| the arena's numbers; the $3^-$ witness | — | new data |
| reachable deficiency spectrum $\mathfrak{D}(\mathcal{V})$ | — | new question |

*Table 6. Collision map. Every mechanism is recovered standard representation theory reached from the vocabulary side. Direction of arrival is the contribution; the theorems are not new.*

**Proposition 5.1. (minimal orbit generation)** Orbit spans are cyclic submodules $\mathbb{Q}[G]\cdot v$, so the minimal orbit count is the minimal number of module generators. The bound max$\lambda$ $\lceil m_\lambda s_\lambda / \dim \lambda \rceil$ evaluates to 15, binding at $1^+$, and an explicit 15-orbit set spanning all six isotypic channels was found. Hence $\mu = 15$ exactly. Of the 90 orbits, 26 are free of dimension 24; the remainder are proper cyclic quotients.

## 5.1 The integral boundary

# 6. From Reader Channel to Field Equation

Sections 2–4 are statements about a module. This section tests whether they are statements about a dynamics, by putting a field on the same arena and verifying every structural hypothesis rather than assuming it.

## 6.1 The model

## 6.2 Linear invisibility and nonlinear sourcing

| step | $\|P_0\Phi\|$ visible, g = 0 | $\|P_1\Phi\|$ null, g = 0 | $\|P_0\Phi\|$ visible, g = 1 | $\|P_1\Phi\|$ null, g = 1 |
|---|---|---|---|---|
| 1 | 0 | 93.28 | 4.62 | 93.28 |
| 3 | 0 | 74.64 | 11.06 | 75.64 |
| 12 | 0 | 35.60 | 23.82 | 40.85 |

*Table 7. Linear versus nonlinear evolution from a purely reader-null initial field. Exactly zero in the visible channel at every linear step; nonzero from the first nonlinear step.*

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## 6.3 The hidden sector is recoverable

Because $P_1(\text{odd}\cdot\text{odd}) = 0$, the odd sector squares wholly into the even channel, so $P_0(\Phi_-^2) = \Phi_-^2$ identically and the first Euler step is exactly $\Phi_+(dt) = dt \cdot g \cdot \Phi_-(0)^2$. The reader can invert its own observation.

> **Result 6.1. (exact recovery up to sign)** The quantity $P_0\Phi(dt)/(dt\cdot g)$ equals $\Phi_-(0)^2$ exactly, so $|\Phi_-(0)[v]|$ is determined at every one of the 66 sites. What is lost is sign, at 58 nonzero sites, and the loss is constrained rather than free: a naive per-site flip leaves the odd channel and is not an admissible state, while global negation is admissible and gives an identical reading.

> **Scope 6.2. (what Section 6 is not)** There is no spacetime, no Lorentz structure, no $\hbar$, no amplitude, no Born rule, no particle content and no gravity here. $\Phi$ is a function on 66 graph vertices; L and the quadratic term were chosen for equivariance and lowest order, not derived. What is bridged is reader channel to field equation. What is not bridged is any reason from the programme's primitives why $\Phi$, L, $\mathcal{N}$ and R should be nature's state, law, interaction and measurement.

# 7. The Operator Is Not Forced: a 22-Parameter Family

Section 6 chose L. The honest question is upstream of the spectrum: which local equivariant operators does the arena permit, and do the primitive constraints select one? Three outcomes were possible — L forced up to scale, a located small family, or no constraint at all.

## 7.1 The orbital algebra

| distance | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| orbitals | 8 | 44 | 123 | 108 | 25 |

*Table 8. Orbital census by graph distance. Locality radius r restricts the equivariant algebra to the orbitals within distance r.*

| constraints imposed | dim of admissible family |
|---|---|
| equivariant, locality radius 1 | 52 |
| + self-adjoint | 30 |
| + conservation $T\mathbb{1} = 0$ | 44 |
| + self-adjoint and conservation | 22 |

*Table 9. The constraint ladder. L = D − A lies in the 22-dimensional space, verified by reconstructing it from its orbital coefficient vector.*

> **Method note 7.1. (symmetry is a coefficient relation)** Self-adjointness of an equivariant operator is not a property each orbital basis matrix must satisfy. The transpose of an orbital is another orbital, $M^{ot} = M_{(}{}^{ot}{}_{)}$, so symmetry is the coefficient condition $c_O = c_{O^t}$. An earlier pass filtered to individually symmetric generators and returned dimension zero, which is impossible since L is both symmetric and conserving. Recorded because the rule generalizes.

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## 7.2 The 22 are edge orbits

The number is not arbitrary. The arena has exactly 22 edge orbits — G-classes of unordered adjacent pairs, of sizes {6, 6, 12×16, 24×4} — so the arena is not edge-transitive, and the 22-dimensional family is precisely one free weight per edge orbit with the diagonal forced by conservation:

$$(L_w f)(u) = \Sigma_{v\sim u} w_{[u,v]} ( f(u) - f(v) ), \quad w \text{ constant on each } G\text{-edge orbit.}$$

**Theorem 7.2. (the Laplacian is a homogeneity assumption)** $L = D - A$ is the point of the family where all 22 weights are equal. Nothing in equivariance, locality at radius one, self-adjointness or conservation makes them equal. Choosing the standard Laplacian assumes edge-transitivity on an arena that does not have it: "all moves cost the same" is imported, not derived.

**Corollary 7.3. (no canonical dispersion relation)** A dispersion relation $\omega(q)$ requires a momentum label, which comes from characters of a translation group. $\mathrm{Aut}(X) \cong C_2 \times A_4$ is finite and nonabelian and supplies none; the arena is not a ring, torus, Bravais lattice or embedded homogeneous space. What the arena does force is a combinatorial causal bound: any strictly local update propagates at most one arena edge per step, so $\mathrm{supp}\ \Phi_t \subseteq N_t(\mathrm{supp}\ \Phi_0)$. Converting that into a physical velocity requires a length per edge and a duration per update, neither of which is supplied.

# 8. The Gradient Condition: 22 → 7

## 8.1 The potential is forced to be invariant

**Lemma 8.1. (equivariant gradients have invariant potentials)** Let $p$ be a $G$-equivariant antisymmetric price on directed arena edges with $p = d\phi$. Then $d(\phi\circ g - \phi) = 0$, so $\phi\circ g - \phi$ is constant on the connected arena; that constant defines a homomorphism $G \to (\mathbb{Q}, +)$, which is zero for a finite group. Hence $\phi\circ g = \phi$, and $\phi$ is constant on the 8 vertex orbits.

## 8.2 The measurement

| quantity | value |
|---|---|
| vertex orbits | 8, sizes {12, 3, 12, 6, 12, 6, 12, 3} |
| edge orbits | 22 |
| edge orbits reversed by some $g \in G$ | 0 |
| edge orbits inside one vertex orbit | 1 |
| edge orbits joining two vertex orbits | 21 |
| rank $B$ | 7 |
| gradient-compatible subspace | 7 of 22 — removes 15 parameters |

*Table 10. The gradient reduction. The surviving dimension is 8 vertex orbits minus the additive constant.*

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**Theorem 8.2. (the Laplacian is excluded by one edge orbit)** Exactly one of the 22 edge orbits has both endpoints in the same vertex orbit. For it $s(a) = t(a)$, so every invariant potential gives price $\phi_{t(a)} - \phi_{s(a)} = 0$. The unweighted Laplacian assigns that orbit weight 1. Therefore $L = D - A \notin \text{im } B$. The exclusion is local, exact, and witnessed by the single edge class whose endpoints are symmetry-equivalent.

## 8.3 The connectivity control

## 8.4 The eight-row audit

The remaining freedom is a scalar on 8 symmetry classes, so candidate primitives are a finite check rather than a research direction. Continuation count is the sharpest candidate because C1 — every distinction must admit lawful continuation — supplies it directly: for an arena state it is the number of admissible one-token moves, i.e. the arena degree.

| candidate potential $\phi$ | values over the 8 vertex orbits | zero-price edges | connected |
|---|---|---|---|
| continuation count (degree) | [12, 14, 8, 8, 10, 10, 6, 6] | 48 of 300 | yes |
| orbit size | [12, 3, 12, 6, 12, 6, 12, 3] | 144 of 300 | yes |
| stabilizer order | [2, 8, 2, 4, 2, 4, 2, 8] | 144 of 300 | yes |
| 3-cell incidence | [13, 16, 9, 8, 10, 10, 4, 4] | 36 of 300 | yes |
| 4-cell incidence | [109, 143, 55, 56, 68, 65, 26, 27] | 24 of 300 | yes |

*Table 11. Candidate orbit potentials. All five are nonconstant and all preserve connectivity. Five distinct projective points after affine normalization.*

**Result 8.3. (C1 supplies a gradient, but not uniquely)** Continuation count is nonconstant on the eight orbits — five distinct values — and the resulting price law zeroes 3 of the 22 edge orbits (against the 1 forced by the gradient condition alone) while leaving the arena connected. So C1 does generate a gradient on this witness, which is the non-vacuous outcome. But four further candidates do so as well, with distinct projective potentials, and nothing in the primitive constraints selects among them.

That is parameter reduction without predictive selection. A candidate earns a stronger status only by predicting something not used in its construction — a spectrum, a stability property, a propagation rate, a nonlinear coupling, or a closure behaviour. That test has not been run.

# 9. Status, Non-Claims, and Open Edges

The derivation chain established in this paper reads:

$$\textit{arena + symmetry} \Rightarrow \textit{22-parameter local conservative family} \Rightarrow \textit{(gradient law)} \Rightarrow \textit{7 parameters} \Rightarrow \textit{(orbit potential)} \Rightarrow L_w \Rightarrow \textit{spectrum.}$$

The first two arrows are executed and exact. The third is open: five candidates survive and none is forced. The fourth needs a semantic decision that this paper does not make — whether a zero signed price means free transport, zero coupling, or an indistinguishable move are different physical hypotheses.

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## 9.1 What is established

## 9.2 What is not established

That $3^-$ is the only pure realizable type: one construction failed for the other five, which is not a proof of impossibility. Anything about the $2^{90}$ orbit lattice beyond existence: it was sampled, never exhausted, so absence-in-sample is weak. Any physical content for Section 6: no spacetime, metric, amplitude, Born rule, particle content, gravity, or dimensionful constant appears, and none is implied. Any bearing on the integral relation gap problem. And any claim that the primitive constraints determine dynamics — Section 8 is the honest falsification of that, locating the residual freedom at seven parameters and then at a scalar on eight classes.

## 9.3 Provenance

## 9.4 The question the direction bought

Section 5 concedes that the representation theory in this paper is recovered. What the vocabulary-side route produced that the standard route does not naturally ask is Definition 3.2. Representation theory answers which submodules of A exist — all of them, by semisimplicity. It does not answer which are reachable by unions of a specified generating family, and the measured answer here is that reachability is granularity-dependent, with the certification criterion of Theorem 4.1 following from it. That is a question about the declaration lattice rather than the module, and within the scope of one literature pass it appears to lack a standard name. It is a question, not a theorem, and is offered as such.

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